

MARBLEHEAD, OH, USA

WMO: 997293

Lat: 41.544N

Lon: 82.731W

Elev: 190

StdP: 99.06

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.8	N/A	N/A	N/A	N/A	N/A	N/A	13.5	-5.2	12.3	-4.9	3.7	220	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	31.4	N/A	29.6	N/A	27.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	10.8	9.2	-18.7	34.1	4.2	1.3	-21.7	35.1	-24.1	35.9	-26.5	36.7	-29.5	37.6		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-3.3	-2.9	2.5	9.2	15.8	21.2	23.8	22.9	19.6	13.0	6.1	1.4
	DBStd	10.57	6.19	5.99	5.31	4.74	4.52	3.30	2.74	2.34	3.52	4.30	4.84	4.96
	HDD10.0	1516	414	362	243	69	4	0	0	0	18	135	270	
	HDD18.3	3248	672	594	492	276	109	11	1	1	26	174	368	526
	CDD10.0	1818	1	1	9	47	182	336	427	399	287	111	17	3
	CDD18.3	510	0	0	0	3	29	97	169	142	62	9	0	0
	CDH23.3	2754	0	0	0	31	163	501	1107	662	264	25	0	0
	CDH26.7	641	0	0	0	4	41	126	282	115	69	4	0	0
Wind	WSAvg	3.9	4.2	4.1	4.5	4.7	3.8	3.5	3.2	3.3	3.6	3.9	3.6	4.0
Precipitation	PrecAvg	865	45	43	65	74	88	98	93	89	79	54	70	66
	PrecMax	1144	117	107	138	183	131	261	320	252	169	118	166	164
	PrecMin	464	5	0	24	23	18	17	25	18	23	1	6	21
	PrecStd	145	31	25	31	36	29	52	59	48	41	28	40	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	14.7	20.3	27.2	30.4	32.2	33.5	32.0	32.0	25.9	19.6	16.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	11.8	10.9	16.9	23.2	27.4	30.0	31.6	29.3	28.9	23.5	17.0	13.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	8.3	7.4	14.0	20.4	24.9	27.8	29.7	27.7	26.1	21.4	15.3	10.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	4.9	4.2	10.5	17.0	22.6	26.2	28.0	26.5	24.4	19.1	13.4	8.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.4	6.4	6.2	7.2	6.8	6.0	5.9	5.5	5.8	6.2	5.8	5.5
		MCDBR	9.4	10.7	11.2	13.3	10.8	9.1	8.3	7.7	8.7	8.6	9.1	8.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.309	0.333	0.358	0.398	0.434	0.447	0.442	0.432	0.387	0.365	0.326	0.315
	taud		2.395	2.307	2.313	2.256	2.199	2.211	2.245	2.261	2.364	2.403	2.474	2.427
	Ebn at Noon		838	868	886	872	846	833	832	830	847	823	814	797
	Edh at Noon		81	103	115	131	143	141	135	129	109	93	75	73
All-Sky Solar Radiation	RadAvg		1.42	2.25	3.42	4.59	5.43	6.04	6.08	5.34	4.30	2.80	1.77	1.19
	RadStd		0.15	0.31	0.30	0.35	0.45	0.40	0.34	0.20	0.37	0.22	0.26	0.13

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	N/A

Nomenclature: See separate page